Chapter 5: Testing and Evaluation (Modules 4, 5 and 6)

Table 5.33 Summary of the selected test cases for Modules 4, 5 and 6

Type	Section	Test cases	Passed	Failed	Not run
Unit	5.1	18	18	0	0
Functional	5.2	18	17	0	1
Business rules (decision tables)	5.3	24	24	0	0
Integration	5.4	8	8	0	0
Total		68	67	0	1

The cases above are a representative selection from the automated suites of Modules 4, 5 and 6 (about six per module per category). "Not run" counts cases that were blocked by the test device; they are explained in Section 5.5.4.

5.1 Unit Testing

Unit Testing 12: Exercise knowledge base, safety rules and plan generator (Module 4, Personalized Workout Plan Generation)

Testing Objective: To ensure the exercise dataset is imported faithfully, equipment and medical rules decide each exercise as written, warning symptoms stop plan generation, the week has the chosen structure, and energy estimates use the MET formula.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M4-U07	The free-exercise-db dataset is imported with every usable record	exercise_catalog.json (pinned commit f00c92c)	876 records, 871 imported, 5 without instructions dropped	Records 876; imported 871; loaded 871; dropped 5	**Pass**
2	ATC-M4-U10	Equipment exercises are offered only for equipment the user owns	owned equipment {dumbbell}; then none	Every offered exercise needs a dumbbell; none offered without equipment	121 offered with a dumbbell, all dumbbell: true; without equipment: 0	**Pass**
3	ATC-M4-U13	Hypertension excludes high-intensity intervals and names the rule	condition "Hypertension"; HIIT exercise	Excluded; rule MED-HTN	Outcome excluded; rules [MED-HTN]	**Pass**
4	ATC-M4-U16	The week has the chosen number of training days; the rest are rest days	healthy user, 4 days a week	7 days: 4 workouts, 3 rest days	7 days; 4 workouts; 3 rest days	**Pass**
5	ATC-M4-U20	Calories = MET × body weight × active hours	MET 3.8, 70 kg, 30 min active	133 kcal	kcal = 133.00	**Pass**
6	ATC-M4-U23	Warning symptoms stop plan generation and give the reason	readiness: chest pain reported	No workout days; gate needsClearance (SCR-SYMPTOMS)	Workout days 0; gate needsClearance; rules [SCR-SYMPTOMS]	**Pass**

Unit Testing 13: Posture engine, repetition counting and form checks (Module 5, Real-Time Posture Detection and Correction)

Testing Objective: To ensure joint angles and landmark visibility are handled correctly, full repetitions are counted and partial ones rejected, holds are timed only in position, and a correction appears only when a fault persists. Inputs are synthetic landmark frames, so these cases verify the logic, not detection accuracy on real people.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M5-U01	Joint angle at the middle landmark is computed correctly	points (1,0), (0,0), (0,1)	90°	Angle = 90.00°	**Pass**
2	ATC-M5-U02	Landmarks below the visibility threshold are not used	squat frame with visibility 0.3 (threshold 0.5)	No measurement	Measurement: none	**Pass**
3	ATC-M5-U05	Three full squats count three reps	3 squats to 88°	3 reps, 0 partial	Reps 3; partial 0	**Pass**
4	ATC-M5-U06	A half repetition is not counted	1 squat that turns back at 125°	0 reps, 1 partial attempt	Reps 0; partial 1	**Pass**
5	ATC-M5-U09	A plank hold is timed only while in position	5 s straight plank, then 3 s sagging	About 5 s held	Held 4 s	**Pass**
6	ATC-M5-U13	A correction is shown only after the fault persists for 3 frames	forward lean of 75° held during the descent	"Keep your back straight" appears, never on the first faulty frame	First correction at frame 26 of 39	**Pass**

Unit Testing 14: Intent recognition, coach replies and voice session control (Module 6, Voice Assistant)

Testing Objective: To ensure transcribed requests are mapped to the right intent (including pain reports), replies use the live session data, and microphone, recognition and closing are handled safely. A simulated recogniser is used, so these cases do not validate speech-recognition accuracy.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M6-U01	A rep question is recognised	"How many reps have I done?"	Intent repCount	Intent = repCount	**Pass**
2	ATC-M6-U06	A pain report is recognised as a safety intent	"My knee hurts"	Intent painOrSymptom	Intent = painOrSymptom	**Pass**
3	ATC-M6-U07	The reply uses the live rep counter, never an invented number	camera session with 2 counted squats; "how many reps"	Reply "You've completed 2 reps…"	Reply: "You've completed 2 reps of 36, 34 to go."	**Pass**
4	ATC-M6-U10	Microphone denied: the assistant never listens	microphone permission denied	State unavailable; 0 listen calls	State unavailable; listen calls 0; message "Microphone access is needed for voice commands. You can still tap a suggestion below."	**Pass**
5	ATC-M6-U12	Recognition errors give a clear message and nothing is spoken	no-match error; then timeout error	"didn't catch that"; "didn't hear anything"; no speech output	No match: "I didn't catch that. Tap the mic and try again."; timeout: "I didn't hear anything. Tap the mic and try again."; spoken 0	**Pass**
6	ATC-M6-U14	Closing releases the microphone and ignores a pending command	listening; close; late "pause workout"	Recognizer cancelled once; session still active	Cancel calls 1; session active	**Pass**

5.2 Functional Testing

Functional Testing 15: Workout plan and exercise detail screens (Module 4, Personalized Workout Plan Generation)

Objective: to ensure the screens show the correct content and react correctly to the user's actions. Cases ATC-M*-F were run headless with Flutter test (widget tests); cases ATC-DEV-* were run on the Android phone with Flutter Integration Test, which operates the real application automatically.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M4-F03	Exercise detail shows the prescribed dose and a start/end illustration	Planned "Jumping Jacks": 2 sets of 30 s, rest 40 s	Sets, per-set time and rest shown; illustration caption shown	Sets 1; "30s" 1; rest "40s" 1; illustration caption 1	**Pass**
2	ATC-M4-F05	"Start exercise" opens mode selection linked to the planned exercise	Generated plan, Monday, 4th exercise	Mode selection opens with plan link (Monday, entry 3)	Opened Push-Ups; plan day 0; entry 3	**Pass**
3	ATC-DEV-M4-01	The Workout tab shows the weekly plan generated for the profile	Test user: healthy, 6 training days a week; open the Workout tab	"Your workout plan", "<intensity> intensity · 6 days a week", 7 day buttons of which 1 is a rest day	Summary "Moderate intensity · 6 days a week" shown 1; day buttons 7; rest-day buttons 1	**Pass**
4	ATC-DEV-M4-03	An exercise opens its detail screen with the planned sets, reps or time and rest, and Back returns to the plan	Workout tab; tap the first exercise card; then Back	Exercise Detail shows the exercise name and the planned dose; the plan is shown again after Back	"Dynamic Warm-Up" 1 sets × 3m, rest 0 s: name 1, sets 1, reps/time 2, rest 1; back on the plan	**Pass**
5	ATC-DEV-M4-04	Regenerate plan produces a different exercise selection (cloud save unavailable in the test session)	Workout tab; "Regenerate plan" → confirm "Regenerate"	The weekly plan changes; the app reports that the new plan could not be saved to the server (no account in the test)	Plan changed: true; message "New plan generated, but it could not be saved to the server."	**Pass**
6	ATC-DEV-M4-06	A reported warning symptom blocks plan generation and the reason is shown (negative case)	Test user; readiness answer: chest pain; Workout tab	"A doctor's clearance comes first" and rule SCR-SYMPTOMS; no exercise cards	Gate shown; rule SCR-SYMPTOMS shown 1; exercise cards 0	**Pass**

Functional Testing 16: Posture detection, mode selection, camera and summary screens (Module 5, Real-Time Posture Detection and Correction)

Objective: to ensure the screens show the correct content and react correctly to the user's actions. Cases ATC-M*-F were run headless with Flutter test (widget tests); cases ATC-DEV-* were run on the Android phone with Flutter Integration Test, which operates the real application automatically.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M5-F05	Camera permission is not requested before the consent toggle is on	Bodyweight Squat; Continue tapped with consent off	Both modes shown; 0 permission requests; consent prompt shown	Mode options 2; camera requests 0; camera screen opened 0	**Pass**
2	ATC-M5-F08	Camera screen counts reps from pose frames and shows status and privacy notice	2 squats fed as pose frames	Reps 2; "Landmarks detected", "Angles calculated"; local-processing notice	Reps 2; status chips 1/1; notice 1	**Pass**
3	ATC-M5-F09	"Stop Session" ends the session and shows the summary with score, stats and feedback	2 good squats, then Stop Session	Summary: "Session Complete!", posture score 100%, Total Reps, Accuracy, form feedback; session saved	Summary 1; score 100%; stats 1/1; feedback 1; saved 1	**Pass**

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
4	ATC-DEV-M5-03	My plan shows every day of the complete plan with each entry's camera capability	Posture Detection → "My plan" → each training day	For every training day: its entries with a capability label	6 training days checked. Full-Body Fitness: 9 entries, capability label 1; Cardio & Core: 8 entries, capability label 1; Strength Mix: 10 entries, capability label 1; Full-Body Fitness: 9 entries, capability label 1; Cardio & Core: 8 entries, capability label 1; Strength Mix: 10 entries, capability label 1	**Pass**
5	ATC-DEV-M5-05	A planned camera exercise opens mode selection; Continue without consent does not start the camera	My plan → a camera-tracked planned exercise → Continue with the consent toggle off	Both modes and the consent switch; the consent message; no camera screen	Entry 0-1: modes 2, consent switch 1, consent message shown, camera screen 0	**Pass**
6	ATC-DEV-M5-06	With consent the real camera session starts on the phone, and Stop Session ends it with a summary	Planned camera exercise → consent on → Continue → about 8 s of live camera → "Stop Session"	Live camera overlay running, exercise name or setup hint, privacy notice; after Stop: "Session Summary" whose reps equal the session counter; saved in memory only	Blocked: camera permission for the installed test build is "denied" on the phone (a reinstall resets it); the tests do not grant permissions. Allow Camera for Baymax in Android Settings and rerun	**Blocked**

Functional Testing 17: Voice assistant screen (Module 6, Voice Assistant)

Objective: to ensure the screens show the correct content and react correctly to the user's actions. Cases ATC-M*-F were run headless with Flutter test (widget tests); cases ATC-DEV-* were run on the Android phone with Flutter Integration Test, which operates the real application automatically.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M6-F02	A spoken question shows the transcript, the reply card, at most three chips and is spoken aloud	Recognised speech "how do I do a squat" (simulated recogniser)	Transcript shown; reply about Bodyweight Squat spoken; ≤ 3 chips	Transcript 1; spoken "Here is how to do Bodyweight Squat"; chips 2	**Pass**
2	ATC-M6-F03	A refused microphone is explained and a retry is offered	Microphone permission denied	"Voice input unavailable", the reason and "Try again"; not listening	Unavailable 1; reason 1; retry 1; listen calls 0	**Pass**
3	ATC-DEV-M6-01	The chatbot's mic opens the Module 6 voice assistant, which shows the state matching the phone's microphone permission	Home → AI chatbot (Module 13) → "Voice assistant"; microphone permission as set on the phone	Voice assistant screen (not the chatbot); granted → "Listening..."; not granted → "Voice input unavailable" with the reason and "Try again"	Phone microphone permission: denied; assistant state: unavailable; reason shown 1; "Try again" 1; chatbot still on top: 0	**Pass**
4	ATC-DEV-M6-02	A quick-action chip is answered from the real session state and spoken with the phone's text-to-speech	No active workout; tap the chip "How many reps?"	Reply card says there is no active workout session (no invented numbers); text-to-speech reports no failure; at most 3 chips	Reply: "There's no active workout session right now. Open an exercise and tap Start Exercise to begin."; engine On-device rules (Dialogflow unavailable); speech output failed false; chips 2	**Pass**
5	ATC-DEV-M6-03	"What's my workout today?" is answered from the personalised plan	Test user's plan; tap the chip "What's my workout today?"	Reply names today's plan day, or says today is a rest day, matching the plan	Plan today: rest day; reply "Today is a rest day in your plan. Light walking or stretching is fine if you feel like moving."	**Pass**
6	ATC-DEV-M6-04	The close button leaves the voice assistant and the app stays usable	Home → chatbot → voice assistant → "Close voice assistant" → Back	Voice assistant closed; chatbot shown; Back returns to Home	Voice assistant closed; chatbot shown 1; Home shown after Back	**Pass**

5.3 Business Rules Testing

Business Rule Testing 14: DT-M4-1 Screening gate (may a plan be generated?)

Table 5.34 Decision table: DT-M4-1 Screening gate (may a plan be generated?)

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Warning symptoms	No	Yes	No	No	No
Heart condition	No	No	Yes	Yes	No
Doctor's clearance	Not needed	Not needed	No	Yes	Not needed
Weight recorded	Yes	Yes	Yes	Yes	No
Action / expected result	Generate plan	Block plan: needsClearance (SCR-SYMPTOMS)	Block plan: needsClearance (SCR-CARDIAC-CLEARANCE)	Generate plan	Block plan: needsInfo (SCR-MISSING-WEIGHT)

Table 5.35 Test cases for DT-M4-1

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M4-B01	Rule 1: Screening gate: symptoms No, heart condition No, clearance Not needed, profile data Yes	Conditions none; symptoms none; clearance null; weight 70 kg	Plan generated	Gate eligible; workout days 4	**Pass**
2	ATC-M4-B02	Rule 2: Screening gate: symptoms Yes, heart condition No, clearance Not needed, profile data Yes	Conditions none; symptoms chest_pain; clearance null; weight 70 kg	No plan; gate needsClearance (SCR-SYMPTOMS)	Gate needsClearance (SCR-SYMPTOMS); workout days 0	**Pass**
3	ATC-M4-B03	Rule 3: Screening gate: symptoms No, heart condition Yes, clearance No, profile data Yes	Conditions Heart Disease; symptoms none; clearance false; weight 70 kg	No plan; gate needsClearance (SCR-CARDIAC-CLEARANCE)	Gate needsClearance (SCR-CARDIAC-CLEARANCE); workout days 0	**Pass**
4	ATC-M4-B04	Rule 4: Screening gate: symptoms No, heart condition Yes, clearance Yes, profile data Yes	Conditions Heart Disease; symptoms none; clearance true; weight 70 kg	Plan generated	Gate eligible; workout days 4	**Pass**
5	ATC-M4-B05	Rule 5: Screening gate: symptoms No, heart condition No, clearance Not needed, profile data No	Conditions none; symptoms none; clearance null; weight 0 kg	No plan; gate needsInfo (SCR-MISSING-WEIGHT)	Gate needsInfo (SCR-MISSING-WEIGHT); workout days 0	**Pass**

Business Rule Testing 15: DT-M4-2 Exercise eligibility (rules checked in order: impact, tags, intensity, injured joint)

Table 5.36 Decision table: DT-M4-2 Exercise eligibility (rules checked in order: impact, tags, intensity, injured joint)

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
Health profile	Healthy	Hypertension	Obese	Healthy	Healthy	Hypertension
Exercise impact	Low	Low	High	High	Low	Low
Restricted tag	None	HIIT	None	None	None	None
Injured joint loaded	No	No	No	Knee	Knee	No
MET recorded	Yes	Yes	Yes	Yes	Yes	No
Action / expected result	eligible	excluded	excluded	excluded	eligibleWithModifications	requiresInformation

Table 5.37 Test cases for DT-M4-2

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M4-B06	Rule 1: Exercise eligibility: Healthy, impact Low, tag None, injury No, MET Yes	Profile Healthy; exercise impact low, tags none, MET 4.0; injuries none	Outcome eligible	Outcome eligible	**Pass**
2	ATC-M4-B07	Rule 2: Exercise eligibility: Hypertension, impact Low, tag HIIT, injury No, MET Yes	Profile At-Risk; exercise impact low, tags hiit, MET 4.0; injuries none	Outcome excluded	Outcome excluded; rules MED-HTN	**Pass**
3	ATC-M4-B08	Rule 3: Exercise eligibility: Obese, impact High, tag None, injury No, MET Yes	Profile Obese; exercise impact high, tags none, MET 4.0; injuries none	Outcome excluded	Outcome excluded; rules BODY-OBESE	**Pass**
4	ATC-M4-B09	Rule 4: Exercise eligibility: Healthy, impact High, tag None, injury Knee, MET Yes	Profile Healthy; exercise impact high, tags none, MET 4.0; injuries knee	Outcome excluded	Outcome excluded; rules INJ-KNEE	**Pass**
5	ATC-M4-B10	Rule 5: Exercise eligibility: Healthy, impact Low, tag None, injury Knee, MET Yes	Profile Healthy; exercise impact low, tags none, MET 4.0; injuries knee	Outcome eligibleWithModifications	Outcome eligibleWithModifications; rules INJ-KNEE	**Pass**
6	ATC-M4-B11	Rule 6: Exercise eligibility: Hypertension, impact Low, tag None, injury No, MET No	Profile At-Risk; exercise impact low, tags none, MET 0.0; injuries none	Outcome requiresInformation	Outcome requiresInformation; rules PRF-AT-RISK	**Pass**

Business Rule Testing 16: DT-M5-2 Repetition acceptance

Table 5.38 Decision table: DT-M5-2 Repetition acceptance

Conditions	Rule 1	Rule 2	Rule 3
Turn-around depth reached	Yes	No	Yes
Duration ≥ 600 ms	Yes	Yes	No
Action / expected result	Count rep	Partial attempt (lowers score)	Ignore as noise

Table 5.39 Test cases for DT-M5-2

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M5-B04	Rule 1: Rep acceptance: depth reached Yes, duration ≥ minimum Yes	One squat (to 88°, 1.8 s)	1 rep, 0 partial	Reps 1; partial 0	**Pass**
2	ATC-M5-B05	Rule 2: Rep acceptance: depth reached No, duration ≥ minimum Yes	One squat (turning at 125°, 1.8 s)	0 rep, 1 partial	Reps 0; partial 1	**Pass**
3	ATC-M5-B06	Rule 3: Rep acceptance: depth reached Yes, duration ≥ minimum No	One squat (to 88°, 0.3 s)	0 rep, 0 partial	Reps 0; partial 0	**Pass**

Business Rule Testing 17: DT-M5-4 Correction feedback trigger

Table 5.40 Decision table: DT-M5-4 Correction feedback trigger

Conditions	Rule 1	Rule 2	Rule 3
Form fault present	No	Yes	Yes
Persists ≥ 3 frames	n/a	No	Yes
Action / expected result	No correction	No correction	Show and speak correction

Table 5.41 Test cases for DT-M5-4

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M5-B09	Rule 1: Correction trigger: fault No, persists ≥ 3 frames n/a	Descent with 10° forward lean for 600 ms (limit 60°)	No correction	Frames with the correction: 0 of 45	**Pass**
2	ATC-M5-B10	Rule 2: Correction trigger: fault Yes, persists ≥ 3 frames No	Descent with 75° forward lean for 66 ms (limit 60°)	No correction	Frames with the correction: 0 of 29	**Pass**
3	ATC-M5-B11	Rule 3: Correction trigger: fault Yes, persists ≥ 3 frames Yes	Descent with 75° forward lean for 600 ms (limit 60°)	"Keep your back straight" shown	Frames with the correction: 16 of 45	**Pass**

Business Rule Testing 18: DT-M6-1 Voice session control

Table 5.42 Decision table: DT-M6-1 Voice session control

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Command	pause workout	resume workout	pause workout	my knee hurts
Session state	active	paused	none	active
Action / expected result	Session paused; reply "Paused…"	Session active; reply "Resuming…"	Session idle; reply "no active workout session…"	Session paused; reply "Please stop the exercise now…"

Table 5.43 Test cases for DT-M6-1

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M6-B01	Rule 1: Voice command "pause workout" with session active	Recognised text "pause workout"; session active	Session paused; reply contains "Paused"	Session paused; reply "Paused. Say resume when you are ready."	**Pass**
2	ATC-M6-B02	Rule 2: Voice command "resume workout" with session paused	Recognised text "resume workout"; session paused	Session active; reply contains "Resuming"	Session active; reply "Resuming. Let's go."	**Pass**
3	ATC-M6-B03	Rule 3: Voice command "pause workout" with session none	Recognised text "pause workout"; session none	Session idle; reply contains "no active workout session"	Session idle; reply "There's no active workout session right now. Open an exercise and tap Start Exercise to begin."	**Pass**
4	ATC-M6-B04	Rule 4: Voice command "my knee hurts" with session active	Recognised text "my knee hurts"; session active	Session paused; reply contains "Please stop the exercise now"	Session paused; reply "Please stop the exercise now and rest. Pain, dizziness or chest discomfort are signals to stop, not to push through. I can't diagnose injuries — if it is severe or doesn't settle, contact a doctor or emergency services. I've paused your session."	**Pass**

Business Rule Testing 19: DT-M6-2 Microphone permission

Table 5.44 Decision table: DT-M6-2 Microphone permission

Conditions	Rule 1	Rule 2	Rule 3
Microphone permission	granted	denied	permanently denied
Action / expected result	Listen	Explain, offer Try again	Explain, offer Open Settings

Table 5.45 Test cases for DT-M6-2

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-M6-B05	Rule 1: Microphone permission granted	Assistant opened; microphone granted	State listening; settings link false	State listening; settings link false; listen calls 1	Pass
2	ATC-M6-B06	Rule 2: Microphone permission denied	Assistant opened; microphone denied	State unavailable; settings link false	State unavailable; settings link false; listen calls 0	Pass
3	ATC-M6-B07	Rule 3: Microphone permission permanently denied	Assistant opened; microphone permanently denied	State unavailable; settings link true	State unavailable; settings link true; listen calls 0	Pass

5.4 Integration Testing

Integration Testing 5: Module 2 health classification to Module 4 workout plan

Testing Objective: to ensure the health category produced by Module 2 sets the intensity ceiling of the generated plan.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-INT-34	The Module 2 health category sets the intensity ceiling of the generated plan	Profile with "Heart Disease" (classified Cardiac), clearance given	Every scheduled exercise with a MET value is light or moderate	Category Cardiac; 36 exercises checked; vigorous 0	Pass

Integration Testing 6: Module 4 workout plan and Module 5 posture sessions (both directions)

Testing Objective: to ensure a planned exercise opens a posture session with its planned dose, and that only a session that reached its target is recorded as completing the planned exercise.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-INT-36	A planned exercise starts a posture session with the planned dose and a camera configuration	Generated plan, Monday; every entry converted to a session	Sets, reps and seconds carried over; every entry resolves a camera capability	9 entries; dose mismatches 0; 4 with camera tracking, 5 Manual Guidance only	Pass
2	ATC-INT-37	Completing every main exercise of the day marks the plan day done once	Monday plan; each main entry finished in Manual Guidance with all sets	Day marked once in the Module 4 activity record	Marked 1× (2026-10-19 Full-Body Fitness); done today 5 entries	Pass
3	ATC-DEV-INT-01	Home → plan → exercise → Manual Guidance session → summary → the planned exercise is recorded as completed (Modules 4 → 5 → 4)	Test user (6 days a week); Workout tab → training day (today, or the first training day when today is a rest day) → its quickest repetition exercise → Start exercise → Manual Guidance → "Completed ✓" for every set (rest timers run) → Finish → Back to Workout Plan	Summary "Confirmed N reps" with N = sets × reps; target reached; session saved with its plan link; plan progress records the entry as done	Monday (today is a rest day): Bodyweight Squat 3×12 (entry 1, plan link 0/1); summary opened by itself after the last set, label "Confirmed 36 reps" 1; target reached true; saved with plan link true (in memory); plan progress done true	Pass

Integration Testing 7: Module 5 posture detection and Module 6 voice assistant (both directions)

Testing Objective: to ensure posture corrections reach the voice coach and voice commands control the running posture session.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-INT-39	A posture correction from the camera engine is spoken once by the cue announcer	Camera squat session; about 3 s at the bottom with 75° forward lean (limit 60°)	"Keep your back straight" spoken exactly once (rate-limited)	Spoken 1: Keep your back straight — lift your chest	**Pass**
2	ATC-INT-42	A spoken "pause" stops rep counting until "resume"	Camera session; "pause workout"; 2 squats while paused; "resume workout"; 1 squat	Paused then active; frames while paused ignored	After pause: paused; reps shown while paused 0; after resume: active	**Pass**
3	ATC-DEV-INT-03	The voice chip "Pause workout" pauses the running posture session (Module 6 → 5)	Planned exercise → Manual Guidance → "Ask Baymax" → chip "How many reps?" → chip "Pause workout" → close	Session paused; the guidance screen shows "Resume"	Session paused; "Resume" shown 1	**Pass**

Integration Testing 8: End-to-end Modules 2, 4, 5 and 6

Testing Objective: to ensure a profile produces a plan whose exercises are done as sessions, queried by voice and recorded as a completed plan day.

No.	Test Case ID	Test case/Test script	Attribute and Value	Expected Result	Actual Result	Result
1	ATC-INT-44	Profile → plan → posture sessions with voice queries → plan day completed	Healthy profile; Monday plan; each main entry done in Manual Guidance; "how many reps" asked in each	Voice replies match the logged reps; Monday marked done once; every session saved	5 sessions; voice reply mismatches 0; saved 5; day marked 1×	**Pass**

5.5 Automated Testing

5.5.1 Automated Testing Strategy

Types of testing automated. Unit, functional (widget), business-rule (decision-table) and integration tests were automated with Flutter test and run without a device. Real-device UI tests and end-to-end flows were automated with Flutter Integration Test on an Android phone. The complete Flutter test suite of the application was also run as a regression check.

Functionality selected and why. For Module 4, the screening gate (warning symptoms, missing data, doctor's clearance), the medical and equipment restrictions, the weekly plan structure and the MET-based energy estimate were selected, because a mistake there could give an unsafe or wrong plan. For Module 5, joint-angle and landmark-visibility handling, repetition acceptance, hold timing, the timing of corrections, camera consent and the session summary were selected, because users rely on the counts and corrections and the camera must not start without consent. For Module 6, intent recognition including pain reports, permission handling, replies built from the live session (never invented numbers) and voice control of a running

session were selected. Integration cases cover the data flow between the health classification (Module 2), the plan (Module 4), posture sessions (Module 5) and the voice assistant (Module 6). Easy but low-value cases (for example simple labels) were not prioritised.

How automation was performed. Each case has a test-case ID in its name and records its objective, input, expected result and the observed actual result, so the tables in this chapter are taken from real runs. Posture cases feed deterministic synthetic pose-landmark sequences into the real posture engine; voice cases use a simulated speech recogniser and speech output in the headless tests. On the phone, Flutter Integration Test starts the real application and drives it through Flutter's gesture system (taps, scrolling, navigation) with a fictional test user held in memory; nobody signs in, finished sessions are kept in memory, and the phone's permission state is read but never changed, so no real account or health record is touched.

Frameworks and tools. Flutter test (flutter_test) and Flutter Integration Test (integration_test, run with flutter drive and test_driver/integration_test.dart), Flutter 3.47.7 with Dart 3.13.5, and the Android Debug Bridge. No other test framework is used for these modules.

Where the tests were executed. Headless tests ran on the development PC (Windows 11 Home). Real-device tests ran on a Redmi Note 11 (2201117TG), Android 13 (API 33), MIUI V816, connected by USB, on 2026-10-18.

Tests that remained manual and why. Real-person posture accuracy (whether reps and corrections are right for a person exercising in front of the camera) cannot be judged automatically and is validated manually; so far only the squat has been checked live (2026-10-15). Live speech recognition with a real voice and the audible quality of spoken replies need a person speaking and listening, so they are validated manually; the automated device tests use fixed quick-action phrases. Saving to the cloud with a real account was not automated, to avoid changing real user data.

5.5.2 Automation Tools and Environment

Table 5.46 Automation tools and environment for Modules 4, 5 and 6

Tool / Framework	Purpose	Version	Test Type	Platform/Environment
flutter_test (Flutter test framework)	Unit, widget (functional), decision-table and integration test cases	Flutter 3.47.7, Dart 3.13.5	Unit / Functional / Business rules / Integration	Windows 11 Home PC, headless
integration_test (Flutter Integration Test)	Automated UI and end-to-end tests of the installed application on a phone	Flutter 3.47.7 SDK package	Functional / Integration (real device)	Redmi Note 11 (2201117TG), Android 13 (API 33), MIUI V816
flutter drive with test_driver/integration_test.dart	Builds and installs the test build, runs it, collects results and screenshots	Flutter 3.47.7 tools	Test runner (real device)	Windows 11 PC, USB to the phone
Android Debug Bridge (adb)	Device connection and device information	adb 1.0.41	-	Windows 11 PC
On-device components under test	Camera plugin with on-device MediaPipe pose detection, speech_to_text, flutter_tts	as locked in pubspec.lock	Functional / Integration (real device)	Redmi Note 11 (2201117TG), Android 13 (API 33), MIUI V816

5.5.3 Automated Test Cases

Table 5.47 Automated test cases selected for Modules 4, 5 and 6

Test Case ID	Related Requirement	Test Objective	Framework / where run	Expected Result	Actual Result	Status
ATC-M4-U07	M4-SFR1	The free-exercise-db dataset is imported with every usable record	Flutter test (headless)	876 records, 871 imported, 5 without instructions dropped	Records 876; imported 871; loaded 871; dropped 5	Pass
ATC-M4-U10	M4-SFR3	Equipment exercises are offered only for equipment the user owns	Flutter test (headless)	Every offered exercise needs a dumbbell; none offered without equipment	121 offered with a dumbbell, all dumbbell: true; without equipment: 0	Pass
ATC-M4-U13	M4-FR1	Hypertension excludes high-intensity intervals and names the rule	Flutter test (headless)	Excluded; rule MED-HTN	Outcome excluded; rules [MED-HTN]	Pass
ATC-M4-U16	M4-FR5	The week has the chosen number of training days; the rest are rest days	Flutter test (headless)	7 days: 4 workouts, 3 rest days	7 days; 4 workouts; 3 rest days	Pass
ATC-M4-U20	M4-SFR2	Calories = MET × body weight × active hours	Flutter test (headless)	133 kcal	kcal = 133.00	Pass
ATC-M4-U23	M4-SFR3	Warning symptoms stop plan generation and give the reason	Flutter test (headless)	No workout days; gate needsClearance (SCR-SYMPTOMS)	Workout days 0; gate needsClearance; rules [SCR-SYMPTOMS]	Pass
ATC-M5-U01	M5-FR16.1	Joint angle at the middle landmark is computed correctly	Flutter test (headless)	90°	Angle = 90.00°	Pass
ATC-M5-U02	M5-FR16.4	Landmarks below the visibility threshold are not used	Flutter test (headless)	No measurement	Measurement: none	Pass
ATC-M5-U05	M5-FR16.3	Three full squats count three reps	Flutter test (headless)	3 reps, 0 partial	Reps 3; partial 0	Pass
ATC-M5-U06	M5-FR16.3	A half repetition is not counted	Flutter test (headless)	0 reps, 1 partial attempt	Reps 0; partial 1	Pass
ATC-M5-U09	M5-SFR2	A plank hold is timed only while in position	Flutter test (headless)	About 5 s held	Held 4 s	Pass
ATC-M5-U13	M5-FR16.2	A correction is shown only after the fault persists for 3 frames	Flutter test (headless)	"Keep your back straight" appears, never on the first faulty frame	First correction at frame 26 of 39	Pass
ATC-M6-U01	M6-SFR1	A rep question is recognised	Flutter test (headless)	Intent repCount	Intent = repCount	Pass
ATC-M6-U06	M6-SFR1	A pain report is recognised as a safety intent	Flutter test (headless)	Intent painOrSymptom	Intent = painOrSymptom	Pass
ATC-M6-U07	M6-FR19.3	The reply uses the live rep counter, never an invented number	Flutter test (headless)	Reply "You've completed 2 reps…"	Reply: "You've completed 2 reps of 36, 34 to go."	Pass
ATC-M6-U10	M6-FR19.2	Microphone denied: the assistant never listens	Flutter test (headless)	State unavailable; 0 listen calls	State unavailable; listen calls 0; message "Microphone access is needed for voice commands. You can still tap a suggestion below."	Pass
ATC-M6-U12	M6-FR19.1	Recognition errors give a clear message and nothing is spoken	Flutter test (headless)	"didn't catch that"; "didn't hear anything"; no speech output	No match: "I didn't catch that. Tap the mic and try again."; timeout: "I didn't hear anything. Tap the mic and try again."; spoken 0	Pass
ATC-M6-U14	M6-FR19.4	Closing releases the microphone and ignores a pending command	Flutter test (headless)	Recognizer cancelled once; session still active	Cancel calls 1; session active	Pass
ATC-M4-F03	M4-FR6	Exercise detail shows the prescribed dose and a start/end illustration	Flutter test (headless)	Sets, per-set time and rest shown; illustration caption shown	Sets 1; "30s" 1; rest "40s" 1; illustration caption 1	Pass
ATC-M4-F05	M4-SFR4	"Start exercise" opens mode selection linked to the planned exercise	Flutter test (headless)	Mode selection opens with plan link (Monday, entry 3)	Opened Push-Ups; plan day 0; entry 3	Pass
ATC-DEV-M4-01	M4-FR5	The Workout tab shows the weekly plan generated for the profile	Flutter Integration Test (Android phone)	"Your workout plan", "<intensity> intensity · 6 days a week", 7 day buttons of which 1 is a rest day	Summary "Moderate intensity · 6 days a week" shown 1; day buttons 7; rest-day buttons 1	Pass
ATC-DEV-M4-03	M4-FR6	An exercise opens its detail screen with the planned sets, reps or time and rest, and Back returns to the plan	Flutter Integration Test (Android phone)	Exercise Detail shows the exercise name and the planned dose; the plan is shown again after Back	"Dynamic Warm-Up" 1 sets × 3m, rest 0 s: name 1, sets 1, reps/time 2, rest 1; back on the plan	Pass
ATC-DEV-M4-04	M4-FR5	Regenerate plan produces a different exercise selection (cloud save unavailable in the test session)	Flutter Integration Test (Android phone)	The weekly plan changes; the app reports that the new plan could not be saved to the server (no account in the test)	Plan changed: true; message "New plan generated, but it could not be saved to the server."	Pass

Test Case ID	Related Requirement	Test Objective	Framework / where run	Expected Result	Actual Result	Status
ATC-DEV-M4-06	M4-SFR3	A reported warning symptom blocks plan generation and the reason is shown (negative case)	Flutter Integration Test (Android phone)	"A doctor's clearance comes first" and rule SCR-SYMPTOMS; no exercise cards	Gate shown; rule SCR-SYMPTOMS shown 1; exercise cards 0	Pass
ATC-M5-F05	M5-FR15.3	Camera permission is not requested before the consent toggle is on	Flutter test (headless)	Both modes shown; 0 permission requests; consent prompt shown	Mode options 2; camera requests 0; camera screen opened 0	Pass
ATC-M5-F08	M5-FR16.3	Camera screen counts reps from pose frames and shows status and privacy notice	Flutter test (headless)	Reps 2; "Landmarks detected", "Angles calculated"; local-processing notice	Reps 2; status chips 1/1; notice 1	Pass
ATC-M5-F09	M5-FR16.5	"Stop Session" ends the session and shows the summary with score, stats and feedback	Flutter test (headless)	Summary: "Session Complete!", posture score 100%, Total Reps, Accuracy, form feedback; session saved	Summary 1; score 100%; stats 1/1; feedback 1; saved 1	Pass
ATC-DEV-M5-03	M5-SFR3	My plan shows every day of the complete plan with each entry's camera capability	Flutter Integration Test (Android phone)	For every training day: its entries with a capability label	6 training days checked. Full-Body Fitness: 9 entries, capability label 1; Cardio & Core: 8 entries, capability label 1; Strength Mix: 10 entries, capability label 1; Full-Body Fitness: 9 entries, capability label 1; Cardio & Core: 8 entries, capability label 1; Strength Mix: 10 entries, capability label 1	Pass
ATC-DEV-M5-05	M5-FR15.3	A planned camera exercise opens mode selection; Continue without consent does not start the camera	Flutter Integration Test (Android phone)	Both modes and the consent switch; the consent message; no camera screen	Entry 0-1: modes 2, consent switch 1, consent message shown, camera screen 0	Pass
ATC-DEV-M5-06	M5-FR16.5	With consent the real camera session starts on the phone, and Stop Session ends it with a summary	Flutter Integration Test (Android phone)	Live camera overlay running, exercise name or setup hint, privacy notice; after Stop: "Session Summary" whose reps equal the session counter; saved in memory only	Blocked: camera permission for the installed test build is "denied" on the phone (a reinstall resets it); the tests do not grant permissions. Allow Camera for Baymax in Android Settings and rerun	Blocked
ATC-M6-F02	M6-FR19.3	A spoken question shows the transcript, the reply card, at most three chips and is spoken aloud	Flutter test (headless)	Transcript shown; reply about Bodyweight Squat spoken; ≤ 3 chips	Transcript 1; spoken "Here is how to do Bodyweight Squat"; chips 2	Pass
ATC-M6-F03	M6-FR19.2	A refused microphone is explained and a retry is offered	Flutter test (headless)	"Voice input unavailable", the reason and "Try again"; not listening	Unavailable 1; reason 1; retry 1; listen calls 0	Pass
ATC-DEV-M6-01	M6-FR19.1	The chatbot's mic opens the Module 6 voice assistant, which shows the state matching the phone's microphone permission	Flutter Integration Test (Android phone)	Voice assistant screen (not the chatbot); granted → "Listening..."; not granted → "Voice input unavailable" with the reason and "Try again"	Phone microphone permission: denied; assistant state: unavailable; reason shown 1; "Try again" 1; chatbot still on top: 0	Pass
ATC-DEV-M6-02	M6-FR19.3	A quick-action chip is answered from the real session state and spoken with the phone's text-to-speech	Flutter Integration Test (Android phone)	Reply card says there is no active workout session (no invented numbers); text-to-speech reports no failure; at most 3 chips	Reply: "There's no active workout session right now. Open an exercise and tap Start Exercise to begin."; engine On-device rules (Dialogflow unavailable); speech output failed false; chips 2	Pass
ATC-DEV-M6-03	M6-FR19.3	"What's my workout today?" is answered from the personalised plan	Flutter Integration Test (Android phone)	Reply names today's plan day, or says today is a rest day, matching the plan	Plan today: rest day; reply "Today is a rest day in your plan. Light walking or stretching is fine if you feel like moving."	Pass
ATC-DEV-M6-04	M6-FR19.4	The close button leaves the voice assistant and the app stays usable	Flutter Integration Test (Android phone)	Voice assistant closed; chatbot shown; Back returns to Home	Voice assistant closed; chatbot shown 1; Home shown after Back	Pass
ATC-M4-B01	M4-SFR3	Screening gate: symptoms No, heart condition No, clearance Not needed, profile data Yes	Flutter test (headless)	Plan generated	Gate eligible; workout days 4	Pass
ATC-M4-B02	M4-SFR3	Screening gate: symptoms Yes, heart condition No, clearance Not needed, profile data Yes	Flutter test (headless)	No plan; gate needsClearance (SCR-SYMPTOMS)	Gate needsClearance (SCR-SYMPTOMS); workout days 0	Pass
ATC-M4-B03	M4-SFR3	Screening gate: symptoms No, heart condition Yes, clearance No, profile data Yes	Flutter test (headless)	No plan; gate needsClearance (SCR-CARDIAC-CLEARANCE)	Gate needsClearance (SCR-CARDIAC-CLEARANCE); workout days 0	Pass

Test Case ID	Related Requirement	Test Objective	Framework / where run	Expected Result	Actual Result	Status
ATC-M4-B04	M4-SFR3	Screening gate: symptoms No, heart condition Yes, clearance Yes, profile data Yes	Flutter test (headless)	Plan generated	Gate eligible; workout days 4	Pass
ATC-M4-B05	M4-SFR3	Screening gate: symptoms No, heart condition No, clearance Not needed, profile data No	Flutter test (headless)	No plan; gate needsInfo (SCR-MISSING-WEIGHT)	Gate needsInfo (SCR-MISSING-WEIGHT); workout days 0	Pass
ATC-M4-B06	M4-FR1	Exercise eligibility: Healthy, impact Low, tag None, injury No, MET Yes	Flutter test (headless)	Outcome eligible	Outcome eligible	Pass
ATC-M4-B07	M4-FR1	Exercise eligibility: Hypertension, impact Low, tag HIIT, injury No, MET Yes	Flutter test (headless)	Outcome excluded	Outcome excluded; rules MED-HTN	Pass
ATC-M4-B08	M4-FR1	Exercise eligibility: Obese, impact High, tag None, injury No, MET Yes	Flutter test (headless)	Outcome excluded	Outcome excluded; rules BODY-OBESE	Pass
ATC-M4-B09	M4-FR1	Exercise eligibility: Healthy, impact High, tag None, injury Knee, MET Yes	Flutter test (headless)	Outcome excluded	Outcome excluded; rules INJ-KNEE	Pass
ATC-M4-B10	M4-FR1	Exercise eligibility: Healthy, impact Low, tag None, injury Knee, MET Yes	Flutter test (headless)	Outcome eligibleWithModifications	Outcome eligibleWithModifications; rules INJ-KNEE	Pass
ATC-M4-B11	M4-FR1	Exercise eligibility: Hypertension, impact Low, tag None, injury No, MET No	Flutter test (headless)	Outcome requiresInformation	Outcome requiresInformation; rules PRF-AT-RISK	Pass
ATC-M5-B04	M5-FR16.3	Rep acceptance: depth reached Yes, duration ≥ minimum Yes	Flutter test (headless)	1 rep, 0 partial	Reps 1; partial 0	Pass
ATC-M5-B05	M5-FR16.3	Rep acceptance: depth reached No, duration ≥ minimum Yes	Flutter test (headless)	0 rep, 1 partial	Reps 0; partial 1	Pass
ATC-M5-B06	M5-FR16.3	Rep acceptance: depth reached Yes, duration ≥ minimum No	Flutter test (headless)	0 rep, 0 partial	Reps 0; partial 0	Pass
ATC-M5-B09	M5-FR16.2	Correction trigger: fault No, persists ≥ 3 frames n/a	Flutter test (headless)	No correction	Frames with the correction: 0 of 45	Pass
ATC-M5-B10	M5-FR16.2	Correction trigger: fault Yes, persists ≥ 3 frames No	Flutter test (headless)	No correction	Frames with the correction: 0 of 29	Pass
ATC-M5-B11	M5-FR16.2	Correction trigger: fault Yes, persists ≥ 3 frames Yes	Flutter test (headless)	"Keep your back straight" shown	Frames with the correction: 16 of 45	Pass
ATC-M6-B01	M6-SFR3	Voice command "pause workout" with session active	Flutter test (headless)	Session paused; reply contains "Paused"	Session paused; reply "Paused. Say resume when you are ready."	Pass
ATC-M6-B02	M6-SFR3	Voice command "resume workout" with session paused	Flutter test (headless)	Session active; reply contains "Resuming"	Session active; reply "Resuming. Let’s go."	Pass
ATC-M6-B03	M6-SFR3	Voice command "pause workout" with session none	Flutter test (headless)	Session idle; reply contains "no active workout session"	Session idle; reply "There's no active workout session right now. Open an exercise and tap Start Exercise to begin."	Pass
ATC-M6-B04	M6-SFR3	Voice command "my knee hurts" with session active	Flutter test (headless)	Session paused; reply contains "Please stop the exercise now"	Session paused; reply "Please stop the exercise now and rest. Pain, dizziness or chest discomfort are signals to stop, not to push through. I can't diagnose injuries — if it is severe or doesn't settle, contact a doctor or emergency services. I've paused your session."	Pass
ATC-M6-B05	M6-FR19.2	Microphone permission granted	Flutter test (headless)	State listening; settings link false	State listening; settings link false; listen calls 1	Pass
ATC-M6-B06	M6-FR19.2	Microphone permission denied	Flutter test (headless)	State unavailable; settings link false	State unavailable; settings link false; listen calls 0	Pass
ATC-M6-B07	M6-FR19.2	Microphone permission permanently denied	Flutter test (headless)	State unavailable; settings link true	State unavailable; settings link true; listen calls 0	Pass
ATC-INT-34	M4-FR2	The Module 2 health category sets the intensity ceiling of the generated plan	Flutter test (headless)	Every scheduled exercise with a MET value is light or moderate	Category Cardiac; 36 exercises checked; vigorous 0	Pass

Test Case ID	Related Requirement	Test Objective	Framework / where run	Expected Result	Actual Result	Status
ATC-INT-36	M5-SFR3	A planned exercise starts a posture session with the planned dose and a camera configuration	Flutter test (headless)	Sets, reps and seconds carried over; every entry resolves a camera capability	9 entries; dose mismatches 0; 4 with camera tracking, 5 Manual Guidance only	Pass
ATC-INT-37	M4-SFR4	Completing every main exercise of the day marks the plan day done once	Flutter test (headless)	Day marked once in the Module 4 activity record	Marked 1× (2026-10-19 Full-Body Fitness); done today 5 entries	Pass
ATC-DEV-INT-01	M4-SFR4	Home → plan → exercise → Manual Guidance session → summary → the planned exercise is recorded as completed (Modules 4 → 5 → 4)	Flutter Integration Test (Android phone)	Summary "Confirmed N reps" with N = sets × reps; target reached; session saved with its plan link; plan progress records the entry as done	Monday (today is a rest day): Bodyweight Squat 3×12 (entry 1, plan link 0/1); summary opened by itself after the last set, label "Confirmed 36 reps" 1; target reached true; saved with plan link true (in memory); plan progress done true	Pass
ATC-INT-39	M6-SFR2	A posture correction from the camera engine is spoken once by the cue announcer	Flutter test (headless)	"Keep your back straight" spoken exactly once (rate-limited)	Spoken 1: Keep your back straight — lift your chest	Pass
ATC-INT-42	M6-SFR3	A spoken "pause" stops rep counting until "resume"	Flutter test (headless)	Paused then active; frames while paused ignored	After pause: paused; reps shown while paused 0; after resume: active	Pass
ATC-DEV-INT-03	M6-SFR3	The voice chip "Pause workout" pauses the running posture session (Module 6 → 5)	Flutter Integration Test (Android phone)	Session paused; the guidance screen shows "Resume"	Session paused; "Resume" shown 1	Pass
ATC-INT-44	M4-SFR4	Profile → plan → posture sessions with voice queries → plan day completed	Flutter test (headless)	Voice replies match the logged reps; Monday marked done once; every session saved	5 sessions; voice reply mismatches 0; saved 5; day marked 1×	Pass

5.5.4 Automated Test Execution and Results

Table 5.48 How the selected suites are run

Suite	Command (run from the app folder)	Type
Unit test cases of Modules 4, 5 and 6 (Flutter test, selected cases)	flutter test test/atc/atc_m456_unit_test.dart -r github --name "ATC-M4-U(07|10|13|16|20|23) |ATC-M5-U(01|02|05|06|09|13) |ATC-M6-U(01|06|07|10|12|14) "	Flutter test
Functional (widget) test cases of Modules 4, 5 and 6 (Flutter test, selected cases)	flutter test test/atc/atc_m456_functional_test.dart -r github --name "ATC-M4-F0[35] |ATC-M5-F0[589] |ATC-M6-F0[23] "	Flutter test
Business-rule (decision-table) test cases of Modules 4, 5 and 6 (Flutter test, selected cases)	flutter test test/atc/atc_m456_rules_test.dart -r github --name "ATC-M4-B(0[1-9]|1[01]) |ATC-M5-B(0[4-6]|09|1[01]) |ATC-M6-B0[1-7] "	Flutter test
Integration test cases of Modules 2, 4, 5 and 6 (Flutter test, selected cases)	flutter test test/atc/atc_m456_integration_test.dart -r github --name "ATC-INT-(34|36|37|39|42|44) "	Flutter test
All real-device cases of one module	cd app && bash integration_test/run_on_device.sh integration_test/<module>_test.dart <run-name>	Flutter Integration Test
Complete regression of the application	cd app && flutter test --concurrency=2	Flutter test

Table 5.49 Execution results of the selected test cases

Framework	Automated tests executed	Passed	Failed	Not run
Flutter test (headless), selected cases	55	55	0	0
Flutter Integration Test on the Redmi Note 11 (2201117TG), selected cases	13	12	0	1
Total selected	68	67	0	1
Complete Flutter regression of the application (context)	993	993	0	13 (skipped in source)

The headless results come from the Flutter test run of 2026-10-18 17:31 (all 55 selected cases passed). The real-device results come from the Flutter Integration Test runs of 2026-10-18 on the Redmi Note 11 (2201117TG), Android 13 (API 33), MIUI V816. ATC-DEV-M5-06 could not be executed: the camera permission of the installed test build was "denied" on the phone after a reinstall, and the tests do not grant permissions themselves; these cases are reported as not run and need a rerun after camera access is allowed. The figures below are terminal screenshots of the selected suites.

[Screenshot pending: run "flutter test test/atc/atc_m456_unit_test.dart -r github --name "ATC-M4-U(07|10|13|16|20|23) |ATC-M5-U(01|02|05|06|09|13) |ATC-M6-U(01|06|07|10|12|14) "" in the project terminal and insert the screenshot here.]

Figure 5.2: Unit test cases of Modules 4, 5 and 6 (Flutter test, selected cases) (pending)

[Screenshot pending: run "flutter test test/atc/atc_m456_functional_test.dart -r github --name "ATC-M4-F0[35] |ATC-M5-F0[589] |ATC-M6-F0[23] "" in the project terminal and insert the screenshot here.]

Figure 5.3: Functional (widget) test cases of Modules 4, 5 and 6 (Flutter test, selected cases) (pending)

[Screenshot pending: run "flutter test test/atc/atc_m456_rules_test.dart -r github --name "ATC-M4-B(0[1-9]|1[01]) |ATC-M5-B(0[4-6]|09|1[01]) |ATC-M6-B0[1-7] "" in the project terminal and insert the screenshot here.]

Figure 5.4: Business-rule (decision-table) test cases of Modules 4, 5 and 6 (Flutter test, selected cases) (pending)

[Screenshot pending: run "flutter test test/atc/atc_m456_integration_test.dart -r github --name "ATC-INT-(34|36|37|39|42|44) "" in the project terminal and insert the screenshot here.]

Figure 5.5: Integration test cases of Modules 2, 4, 5 and 6 (Flutter test, selected cases) (pending)

[Screenshot pending: run "Select-String … run6_module4_workout\flutter_drive.log" in the project terminal and insert the screenshot here.]

Figure 5.6: Recorded real-device run of Module 4 (Flutter Integration Test on the Redmi Note 11, log of 2026-10-18) (pending)

[Screenshot pending: run "Select-String … run7_module5_posture\flutter_drive.log" in the project terminal and insert the screenshot here.]

Figure 5.7: Recorded real-device run of Module 5 (Flutter Integration Test on the Redmi Note 11, log of 2026-10-18) (pending)

[Screenshot pending: run "Select-String … run6_module6_voice and run7_cross_module\flutter_drive.log" in the project terminal and insert the screenshot here.]

Figure 5.8: Recorded real-device runs of Module 6 and the cross-module flows (Flutter Integration Test, logs of 2026-10-18) (pending)

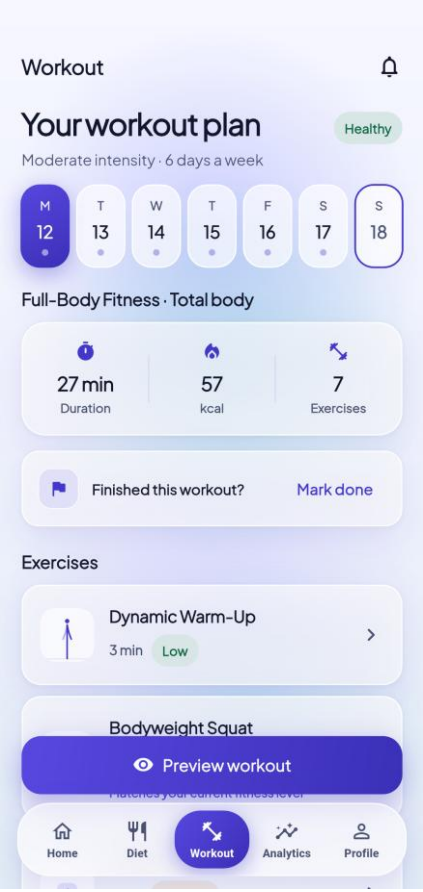


Figure 5.9: Screen captured by ATC-DEV-INT-01 on the Redmi Note 11 at the end of the case (Flutter Integration Test)

5.5.5 Requirement Traceability for Automated Testing

Table 5.50 Requirement Traceability Matrix for Modules 4, 5 and 6 (selected cases)

Requirement ID	Requirement Description	Use Case / Feature	Automated Test Case ID	Automation Tool / Framework	Automated (Y/N)	Status
M4-FR1	Apply medical restriction rules based on health classification before generating a plan	Module 4 Workout Plan; Business rule, Unit	ATC-M4-U13, ATC-M4-B06, ATC-M4-B07, ATC-M4-B08, ATC-M4-B09, ATC-M4-B10, ATC-M4-B11	Flutter test	Yes	Pass (7/7)
M4-FR2	Select workout template appropriate to the health category	Module 4 Workout Plan; Integration	ATC-INT-34	Flutter test	Yes	Pass (1/1)

Requirement ID	Requirement Description	Use Case / Feature	Automated Test Case ID	Automation Tool / Framework	Automated (Y/N)	Status
M4-FR5	Generate a weekly workout schedule with exercises, sets, reps and intensity badges	Module 4 Workout Plan; Functional (real device), Unit	ATC-M4-U16, ATC-DEV-M4-01, ATC-DEV-M4-04	Flutter Integration Test; Flutter test	Yes	Pass (3/3)
M4-FR6	Exercise Detail screen with muscles, metrics and technique steps	Module 4 Workout Plan; Functional, Functional (real device)	ATC-M4-F03, ATC-DEV-M4-03	Flutter Integration Test; Flutter test	Yes	Pass (2/2)
M4-SFR1	Import the exercise dataset with stable, unique IDs (added during implementation)	Module 4 Workout Plan; Unit	ATC-M4-U07	Flutter test	Yes	Pass (1/1)
M4-SFR2	Estimate energy from Compendium MET values: MET x weight x time (added during implementation)	Module 4 Workout Plan; Unit	ATC-M4-U20	Flutter test	Yes	Pass (1/1)
M4-SFR3	Screen readiness (warning symptoms, clearance, missing data) and owned equipment before planning (added during implementation)	Module 4 Workout Plan; Business rule, Functional (real device), Unit	ATC-M4-U10, ATC-M4-U23, ATC-DEV-M4-06, ATC-M4-B01, ATC-M4-B02, ATC-M4-B03, ATC-M4-B04, ATC-M4-B05	Flutter Integration Test; Flutter test	Yes	Pass (8/8)
M4-SFR4	Record completion of planned exercises from posture sessions (added during implementation)	Module 4 Workout Plan; Functional, Integration, Integration (real device)	ATC-M4-F05, ATC-INT-37, ATC-DEV-INT-01, ATC-INT-44	Flutter Integration Test; Flutter test	Yes	Pass (4/4)
M5-FR15.3	The system shall display a camera access consent toggle with a privacy notice before enabling camera-based posture detection.	Module 5 Posture Detection; Functional, Functional (real device)	ATC-M5-F05, ATC-DEV-M5-05	Flutter Integration Test; Flutter test	Yes	Pass (2/2)
M5-FR16.1	The system shall display a live camera feed with a skeletal landmark overlay showing detected body joints and angle measurements.	Module 5 Posture Detection; Unit	ATC-M5-U01	Flutter test	Yes	Pass (1/1)
M5-FR16.2	The system shall display a real-time posture correction feedback banner (e.g., 'Keep your back straight') during camera detection sessions.	Module 5 Posture Detection; Business rule, Unit	ATC-M5-U13, ATC-M5-B09, ATC-M5-B10, ATC-M5-B11	Flutter test	Yes	Pass (4/4)
M5-FR16.3	The system shall display a live rep counter on the camera detection screen that increments as the system detects completed repetitions.	Module 5 Posture Detection; Business rule, Functional, Unit	ATC-M5-U05, ATC-M5-U06, ATC-M5-F08, ATC-M5-B04, ATC-M5-B05, ATC-M5-B06	Flutter test	Yes	Pass (6/6)
M5-FR16.4	The system shall display status indicators confirming 'Landmarks detected' and 'Angles calculated' during an active posture session.	Module 5 Posture Detection; Unit	ATC-M5-U02	Flutter test	Yes	Pass (1/1)
M5-FR16.5	The system shall provide a 'Stop Session' button that ends the camera detection session and navigates to the Session Summary screen.	Module 5 Posture Detection; Functional, Functional (real device)	ATC-M5-F09, ATC-DEV-M5-06	Flutter Integration Test; Flutter test	Yes	Pass (1/1); not run (1, blocked on device)
M5-SFR2	Time static holds such as the plank in camera mode (added during implementation)	Module 5 Posture Detection; Unit	ATC-M5-U09	Flutter test	Yes	Pass (1/1)
M5-SFR3	Start posture sessions from the personalised plan as well as the library (added during implementation)	Module 5 Posture Detection; Functional (real device), Integration	ATC-DEV-M5-03, ATC-INT-36	Flutter Integration Test; Flutter test	Yes	Pass (2/2)
M6-FR19.1	The system shall display a full-screen voice command interface with an animated audio waveform and a 'Listening...' status label when activated.	Module 6 Voice Assistant; Functional (real device), Unit	ATC-M6-U12, ATC-DEV-M6-01	Flutter Integration Test; Flutter test	Yes	Pass (2/2)
M6-FR19.2	The system shall display a central microphone button that toggles the voice listening state on or off.	Module 6 Voice Assistant; Business rule, Functional, Unit	ATC-M6-U10, ATC-M6-F03, ATC-M6-B05, ATC-M6-B06, ATC-M6-B07	Flutter test	Yes	Pass (5/5)
M6-FR19.3	The system shall display Baymax's voice response in a bottom card panel with the response text and contextual quick-action buttons (e.g., 'Start workout', 'Lower intensity', 'Show exercises').	Module 6 Voice Assistant; Functional, Functional (real device), Unit	ATC-M6-U07, ATC-M6-F02, ATC-DEV-M6-02, ATC-DEV-M6-03	Flutter Integration Test; Flutter test	Yes	Pass (4/4)
M6-FR19.4	The system shall provide a close (×) button to exit the voice command interface without executing any command.	Module 6 Voice Assistant; Functional (real device), Unit	ATC-M6-U14, ATC-DEV-M6-04	Flutter Integration Test; Flutter test	Yes	Pass (2/2)
M6-SFR1	Recognise workout intents from the transcribed speech (added during implementation)	Module 6 Voice Assistant; Unit	ATC-M6-U01, ATC-M6-U06	Flutter test	Yes	Pass (2/2)
M6-SFR2	Speak posture corrections and answer form questions from the live session (added during implementation)	Module 6 Voice Assistant; Integration	ATC-INT-39	Flutter test	Yes	Pass (1/1)
M6-SFR3	Control the posture session by voice: pause, resume, stop with confirmation (added during implementation)	Module 6 Voice Assistant; Business rule, Integration, Integration (real device)	ATC-M6-B01, ATC-M6-B02, ATC-M6-B03, ATC-M6-B04, ATC-INT-42, ATC-DEV-INT-03	Flutter Integration Test; Flutter test	Yes	Pass (6/6)